

Low-Resource Filipino Adaptation of OmniVoice for Voice Cloning and Text-to-Speech

Research Project Proposal

Current Trends

May 20, 2026

Research Problem

Voice cloning is an emerging AI technology that can synthesize speech in a target speaker's voice from a short reference audio.

- Most high-quality voice cloning systems are strongest in high-resource languages.
- Filipino speech AI remains relatively low-resource.
- There are few open, Filipino-focused voice cloning workflows.
- A practical opportunity is adapting an existing multilingual model instead of training from scratch.

Motivation

- Filipino voice cloning can support education, accessibility, narration, localization, and language learning tools.
- Multilingual TTS models report wide language coverage, but quality can still vary across low-resource languages.
- OmniVoice supports Filipino, but its language table lists only about **7.71 hours** of Filipino in the training mixture.

Research Opportunity

Evaluate whether curated Filipino speech can improve a pretrained multilingual voice cloning model.

Background: OmniVoice

OmniVoice: Towards Omnilingual Zero-Shot Text-to-Speech with Diffusion Language Models

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Abstract

We present OmniVoice, a monolingual zero-shot text-to-speech (TTS) model that scales to over 600 languages. At its core is a novel diffusion language model-style discrete variational auto-regressive (NAR) architecture. Unlike conventional discrete NAR models that suffer from performance bottlenecks in complex two-stage (text-to-acoustic to acoustic) pipelines, OmniVoice directly maps text to acoustic representations. This simplified approach is facilitated by two key technical innovations: (1) a full-codebook random masking strategy for efficient training, and (2) initialization from a pre-trained LLM to ensure superior intelligibility. By leveraging a 500k-hour multilingual dataset created entirely from open-source data, OmniVoice achieves the broadest language coverage to date and delivers state-of-the-art performance across Chinese, English, and diverse multilingual benchmarks. Our code and pre-trained models are publicly available¹.

1 Introduction

Zero-shot text-to-speech (TTS) models trained on large-scale datasets have demonstrated a remarkable ability to generate high-quality speech conditioned on only a few seconds of reference audio [1, 2]. Despite these advances, most existing models support only a limited set of languages, often leaving hundreds of low-resource languages behind. Expanding language coverage is not merely a technical challenge but also a crucial step toward creating speech technologies in languages across the globe. To address this gap, we aim to develop a monolingual zero-shot TTS model supporting hundreds of languages. Realizing such extensive scaling requires a TTS architecture with exceptional capacity and robustness to handle diverse linguistic patterns.

In the pursuit of such scalable and high-quality TTS, current research primarily adheres to two paradigms: autoregressive (AR) [3, 4, 5, 6, 7, 8, 9, 10, 11] and non-autoregressive (NAR) models [12, 13, 14]. NAR models offer advantages in both inference speed (via parallel decoding) and robustness (owing to bidirectional context) [12, 14, 15]. Within NAR frameworks, models can be broadly categorized into continuous latent-based [14, 16, 17, 18] and discrete-token based [19]. The latter was shown to yield superior prosodic diversity [20].

However, state-of-the-art (SOTA) discrete-token NAR systems [19] typically rely on complex two-stage cascaded pipelines (text-to-acoustic followed by acoustic-to-acoustic). While such decoupling simplifies the training of individual modules, it also introduces significant drawbacks: (1) error propagation, where inaccuracies in acoustic predictions degrade the final audio quality; and (2) information bottlenecks, where low-strate acoustic representations sacrifice fine-grained acoustic details. While single-stage alternatives [21] attempt to bypass these issues, they have historically lagged behind two-stage systems in terms of speech intelligibility.

¹<https://github.com/k2-fsa/omnivoice>

- Open-source multilingual zero-shot TTS and voice cloning model.
- Supports 600+ languages, including Filipino.
- Uses a diffusion language model-style masked-token generation approach.
- Supports voice cloning, voice design, and automatic voice generation.
- Provides official fine-tuning scripts.

Source: Zhu et al. (2026).

Research Question and Scope

Main Research Question

How much does Filipino-specific fine-tuning improve OmniVoice performance for Filipino TTS and voice cloning across intelligibility, naturalness, and speaker similarity metrics?

- Model: k2-fsa/OmniVoice
- Language: Filipino (fil)
- Dataset: Filipino subset of UP-DSP Philippine Languages Database
- Method: fine-tuning, not training from scratch
- Output: generated samples and evaluation after implementation
- Baseline: measure change against the base OmniVoice checkpoint

Objectives

1. Prepare a clean Filipino speech-text dataset for OmniVoice fine-tuning.
2. Fine-tune the pretrained OmniVoice checkpoint on Filipino read speech.
3. Generate Filipino speech using both base and fine-tuned models.
4. Evaluate intelligibility, naturalness, and speaker similarity.
5. Produce a reproducible workflow and demo samples for the existing voice cloning app.

Related Work

- **VALL-E / Neural Codec Language Models:** zero-shot TTS with discrete speech tokens.
- **Voicebox:** text-guided multilingual speech generation at scale.
- **F5-TTS and E2 TTS:** efficient non-autoregressive zero-shot TTS with flow matching.
- **MaskGCT:** masked generative codec modeling for zero-shot TTS.
- **OmniVoice:** extends multilingual zero-shot TTS toward 600+ languages.
- **UP-DSP PLD:** Filipino and Philippine-language data for low-resource speech systems.

Dataset

UP-DSP Philippine Languages Database



The screenshot shows the Mozilla Data Collective interface for the 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. The header includes the Mozilla Data Collective logo and navigation links: 'Browse Datasets', 'API', 'Contribute Datasets', and 'Data Assistant'. The main title is 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. Below the title, it shows the license 'CC-BY-NC-4.0' and the steward 'UP EEEI - Digital Signal...'. Metadata includes 'Task: ASR', 'Release Date: 3/18/2026', 'Format: WAV, LOG', and 'Size: 45.63 GB'. There are buttons for 'Download' and 'Connect API', and a 'Share' icon. At the bottom, there is a 'Files' section with 'Latest Version' and a 'Description' section stating 'This dataset contains multilingual, text and speech pairs for ten Philippine languages namely Filipino, English, ...'.

Mozilla Data Collective

Browse Datasets API Contribute Datasets Data Assistant

UP - DSP - Philippine Languages Database (UP-DSP-PLD)

License: CC-BY-NC-4.0 Steward: UP EEEI - Digital Signal...

Task: ASR Release Date: 3/18/2026 Format: WAV, LOG Size: 45.63 GB

Download Connect API Share

Files Latest Version

Description This dataset contains multilingual, text and speech pairs for ten Philippine languages namely Filipino, English, ...

- Released through Mozilla Data Collective
- WAV audio with LOG transcript files
- License: CC-BY-NC-4.0
- Research use only; not for commercial deployment

Source: UP EEEI - DSP Laboratory (2026).

Filipino Dataset Statistics

Language	Gender	Speaker Count	Utterance Count	Audio Duration		Tokens	
				Total (h:m:s)	Average (s)	Total	Unique
Bikolano (bik)	F	121	39,260	60:55:58	5.5873	321,721	16,049
	M	85	27,684	40:17:36	5.2397	206,642	14,631
	all	206	66,944	101:13:35	5.4436	528,363	17,005
Cebuano (ceb)	F	86	34,956	35:47:01	3.6852	144,882	8,026
	M	86	27,477	27:44:58	3.6357	114,563	7,267
	all	152	62,433	63:31:59	3.6634	259,445	6,844
English (eng)	F	23	3,156	4:31:30	5.1617	29,376	4,363
	M	7	888	1:01:40	4.1675	8,050	1,483
	all	30	4,044	5:33:11	4.9434	37,426	4,729
Filipino (fil)	F	79	30,617	31:43:55	3.7311	205,346	10,861
	M	56	22,262	20:50:48	3.3712	138,088	7,994
	all	135	52,879	48:56:36	3.5796	343,434	11,481
Hiligaynon (hil)	F	48	17,079	21:49:43	4.6012	99,087	5,397
	M	43	14,908	19:21:58	4.6766	84,676	4,906
	all	91	31,987	41:11:42	4.6363	183,763	5,767
Ilokano (ilo)	F	64	15,429	25:46:37	6.0145	131,316	11,603
	M	60	14,513	25:44:43	6.3862	130,500	11,642
	all	124	29,942	51:31:20	6.1947	261,816	13,270
Kapampangan (pam)	F	104	35,024	49:42:02	5.1086	225,595	12,827
	M	83	26,926	40:37:22	5.4313	176,947	12,629
	all	187	61,950	90:19:25	5.2488	402,542	14,221
Pangasinan (pag)	F	12	3,959	6:01:36	5.4802	24,819	4,302
	M	6	1,945	3:07:00	5.7687	11,698	3,148
	all	18	5,904	9:08:36	5.5793	36,517	4,773
Tausug (tsg)	F	4	1,185	1:43:09	5.2236	12,684	2,023
	M	9	2,103	3:06:34	5.3233	7,279	1,536
	all	13	3,288	4:49:45	5.2874	19,963	2,376
Waray-Waray (war)	F	48	15,337	22:51:12	5.3843	94,764	6,071
	M	26	8,500	12:04:10	5.1118	52,704	5,518
	all	74	23,837	34:55:23	5.2743	147,468	6,291
Total	-	1,030	343,208	454:49:43	4.7708	2,220,737	-

Table 2: Summary statistics for the Philippine Languages Database. Below the each Philippine language name is its language ID in parenthesis, based from the ISO 639-3 standard, as published in Ethnologue (Eberhard et al., 2024). Note that the total token and unique token counts do not include yet the transcripts from the spontaneous speech part as this part of the corpora is still being transcribed.

Filipino subset	Value
Language ID	fil
Speakers	135
Utterances	52,879
Duration	48:56:36
Avg. utterance length	3.58 seconds

Enough for a small fine-tuning study while staying feasible on Kaggle.

Source: Guevara et al. (2024).

Data Choice

Read Speech

- Speaker reads a prepared prompt.
- Transcript matches the spoken sentence.
- Best first subset for TTS fine-tuning.

Spontaneous Speech

- Speaker answers freely.
- Prompt may not be the transcript.
- Requires transcript verification.

Project Decision

Use Filipino read speech first to reduce transcript-audio mismatch.

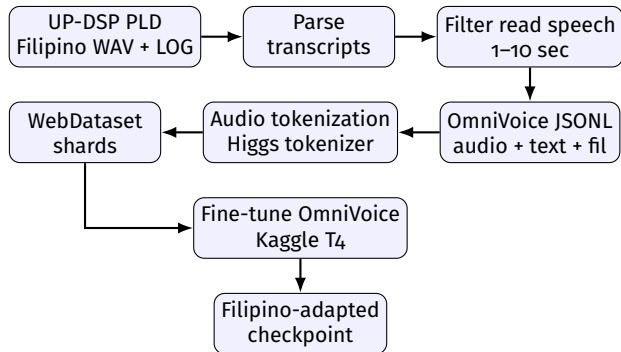
OmniVoice Architecture

1. Text is encoded with a language-model tokenizer.
2. Reference audio is converted into discrete acoustic tokens.
3. The model receives text, language tags, and optional reference voice tokens.
4. During training, some acoustic tokens are masked.
5. The model learns to reconstruct the masked tokens.
6. During inference, generated tokens are decoded into speech audio.

Fine-Tuning Goal

Adapt the text-to-acoustic-token mapping toward Filipino speech patterns.

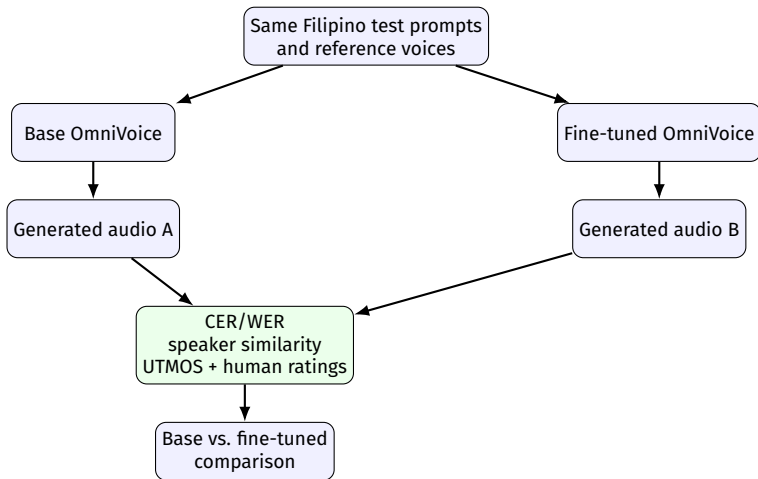
Proposed Fine-Tuning Pipeline



Methodology

1. Acquire the Filipino PLD subset.
2. Parse WAV and LOG files into OmniVoice JSONL format.
3. Filter clean read speech clips between 1 and 10 seconds.
4. Split into train, development, and test sets.
5. Tokenize audio using the OmniVoice audio tokenizer.
6. Fine-tune k2-fsa/OmniVoice on Kaggle T4 GPUs.
7. Generate matched test samples from base and fine-tuned models.
8. Compare outputs using metrics and listener evaluation.

Evaluation Design



Fine-Tuning Setup

- Environment: Kaggle notebook with 2x NVIDIA T4 GPUs.
- Tools: PyTorch, Hugging Face Transformers, Accelerate, DeepSpeed.

Setting	Planned value
Starting checkpoint	k2-fsa/OmniVoice
Data size	5-10 hours clean read speech
Precision	FP16
Attention	SDPA
Learning rate	5e-6
Steps	1,000-2,000
Optimization	DeepSpeed ZeRO-2

Model Testing and Metrics

System	Description
Base OmniVoice	Original pretrained checkpoint
Fine-tuned OmniVoice	Adapted using Filipino read speech

- **CER/WER:** ASR transcript vs. target text.
- **Speaker similarity:** reference vs. generated voice embeddings.
- **UTMOS:** predicted naturalness, if feasible.
- **Human ratings:** pronunciation, naturalness, intelligibility, speaker similarity, preference.

OmniVoice uses the same metric families: WER/CER, SIM-o, UTMOS, CMOS, and SMOS.

How the Metrics Work

- **CER/WER: “Did it say the right words?”**
ASR transcribes generated audio; the transcript is compared with the target text. Lower is better.
- **Speaker similarity: “Does it sound like the reference?”**
Speaker embeddings from reference and generated audio are compared. Higher is better.
- **UTMOS: “Does it sound natural?”**
A prediction model estimates naturalness from the generated waveform. Higher is better.
- **Human ratings: “Which sounds better to Filipino listeners?”**
Raters judge pronunciation, naturalness, intelligibility, similarity, and preference.

Expected Contribution

- Practical Filipino OmniVoice fine-tuning workflow.
- Cleaned Filipino JSONL manifests and experiment setup.
- Base-vs-fine-tuned comparison using measurable metrics.
- Generated audio samples for the existing voice cloning application.
- Evidence on whether low-resource Filipino adaptation is feasible under limited compute.

Application Use Case: Voice Cloning

VOICE CLONE

Upload a short clip with its transcript and save a reusable voice for later generation.

- ☐ Name your voice, upload reference audio, and add the matching transcript.
- ☐ Keep cloned voices organized in one library.
- ☐ Reuse the same saved voice across future generations.

OPEN CLONE →

UPLOAD

RECORD

RECORD REFERENCE AUDIO

0:12

Aim for 10-20 seconds. Recording stops automatically at 60 seconds.

RECORDING...

STOP

CLEAR

VOICE NAME

Aria (warm, close-mic)

TRANSCRIPT

The quick brown fox jumps over the lazy dog near the riverbank.

LANGUAGE

English

TRY EXAMPLE VOICE

CLONE VOICE

Application Use Case: Speech Generation

VOICE

Aria (warm, close-mic) ▾

LANGUAGE

English ▾

TEXT

63 / 100

The quick brown fox jumps over the lazy dog near the riverbank.

GENERATE SPEECH

Speech Generation

PENDING...



DOWNLOAD

GENERATE

Pick a saved voice, paste text, and queue speech generation.

- ☐ The text cap follows the active Qwen runtime settings.
- ☐ Queued jobs keep running in the background until audio is ready.
- ☐ Download finished audio when it sounds right.

OPEN GENERATE →

Application Use Case: Clone Workflow

CLONE ⓘ

UPLOAD RECORD

RECORD REFERENCE AUDIO 0:00

Aim for 10-20 seconds. Recording stops automatically at 60 seconds.

3 2 1

✕

▼ Need something to read?

VOICE NAME

e.g. Duncan (calm, close-mic)...

TRANSCRIPT

Paste the transcript of the reference audio...

0 chars

LANGUAGE

Auto ▼

TRY EXAMPLE VOICE

CLONE VOICE

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